

A Concept White Paper by James Howland (2026)

Abstract

This white paper proposes a distributed optical interferometer spanning the orbits of multiple planets within the Solar System. By deploying a network of space-based telescopes—each comparable to the James Webb Space Telescope (JWST)—into heliocentric orbits and synchronizing their observations using atomic clocks, it becomes possible to synthesize an effective aperture tens of astronomical units (AU) in diameter. Using GPU-accelerated correlation and phase reconstruction, this system could achieve unprecedented angular resolution, enabling direct imaging of exoplanetary surfaces, atmospheric features, and potentially technosignatures. This concept leverages existing technologies in interferometry, atomic timekeeping, and GPU compute, scaled to a solar-system baseline.

1. Introduction

Direct imaging of Earth-like exoplanets remains one of the greatest challenges in astrophysics. Current telescopes lack the aperture size required to resolve surface-level detail on planets orbiting nearby stars. Even next-generation observatories cannot overcome the diffraction limits imposed by their physical size.

This paper introduces a scalable architecture for a Solar-System Interferometric Array (SSIA): a distributed network of synchronized telescopes positioned across planetary orbits. By combining their observations interferometrically, the system synthesizes an aperture tens of AU wide—orders of magnitude larger than any single telescope could ever be.

This approach is physically feasible, technologically grounded, and capable of producing exoplanet images with continent-level resolution.

2. Background and Motivation

2.1 Diffraction Limits

2.2

The angular resolution of a telescope is constrained by:

$$[\ \theta \approx \frac{\lambda}{D}]$$

Where:

(λ) = wavelength

(D) = aperture diameter

To resolve an Earth-sized planet at 4.2 light-years (Proxima b), a telescope would require an aperture comparable to the size of Earth's orbit. No monolithic structure can achieve this.

2.3 Interferometry

2.4

Interferometry allows multiple telescopes to act as a single, larger aperture. This technique is used in:

VLBI (Earth-sized radio telescopes)

ALMA (millimeter interferometry)

Event Horizon Telescope (black hole imaging)

Scaling this concept to the Solar System unlocks unprecedented resolution.

3. Proposed Architecture

3.1 Distributed Telescope Network

3.2

Deploy JWST-class telescopes into heliocentric orbits at:

Earth

Mars

Jupiter

Saturn

Uranus

Neptune

Each telescope maintains:

Cryogenic optics

Infrared and visible sensors

Precision pointing

Onboard atomic clocks

High-bandwidth laser communication

3.3 Atomic Clock Synchronization

3.4

Each observatory uses:

Optical lattice clocks

Femtosecond timing precision

Relativistic corrections

This allows phase-coherent imaging across AU-scale baselines.

3.3 Data Transmission

Observatories transmit raw photon-phase data to Earth using:

Laser communication

Error-corrected encoding

Time-stamped packets

3.5 GPU-Accelerated Correlation

3.6

On Earth, a supercomputer cluster (NVLink-connected GPU nodes) performs:

Cross-correlation

Phase reconstruction

Wavefront synthesis

Interferometric imaging

This is similar to the Event Horizon Telescope pipeline, but scaled massively.

4. Expected Imaging Capabilities

4.1 Angular Resolution

4.2

With a baseline of 30–40 AU:

[$\theta \approx 10^{-13}$ radians]

This enables:

Continent-scale imaging of exoplanets

Detection of oceans and ice caps

Atmospheric weather patterns

Rotational mapping

Seasonal changes

4.3 Potential Technosignature Detection

4.4

The system could detect:

Artificial night-side illumination

Large-scale structures

Atmospheric industrial signatures

5. Feasibility Analysis

5.1 Technological Readiness

5.2

Key components already exist:

JWST-class optics

Atomic clocks

Laser communication

GPU supercomputing

Interferometric algorithms

5.3 Engineering Challenges

5.4

Major challenges include:

Long-baseline phase stability

Orbital station-keeping

Data volume management

Thermal control

Multi-AU communication latency

None of these violate known physics.

6. Mission Concept

A phased deployment strategy:

Phase 1:

Two telescopes in Earth–Mars orbits (baseline: 1.5 AU)

Phase 2:

Add Jupiter and Saturn orbiters (baseline: 10 AU)

Phase 3:

Full Solar-System Array (baseline: 30–40 AU)

Phase 4:

Dedicated GPU supercluster for correlation

7. Scientific Impact

The SSIA would enable:

First-ever surface maps of exoplanets

Atmospheric composition analysis

Climate and weather modeling

Detection of biosignatures

Detection of technosignatures

Unprecedented tests of planetary formation theories

This would be one of the most transformative scientific instruments ever built.

8. Conclusion

A solar-system-scale interferometric telescope is physically possible, technologically grounded, and scientifically revolutionary. By deploying synchronized telescopes across planetary orbits and combining their observations with GPU-accelerated correlation, humanity could directly image exoplanets with unprecedented clarity.

This concept represents a bold but achievable step toward answering one of humanity's oldest questions: What do other worlds truly look like?

9. References